Data Mining
Classification: Alternative Techniques

Lecture Notes for Chapter 4

Instance-Based Learning

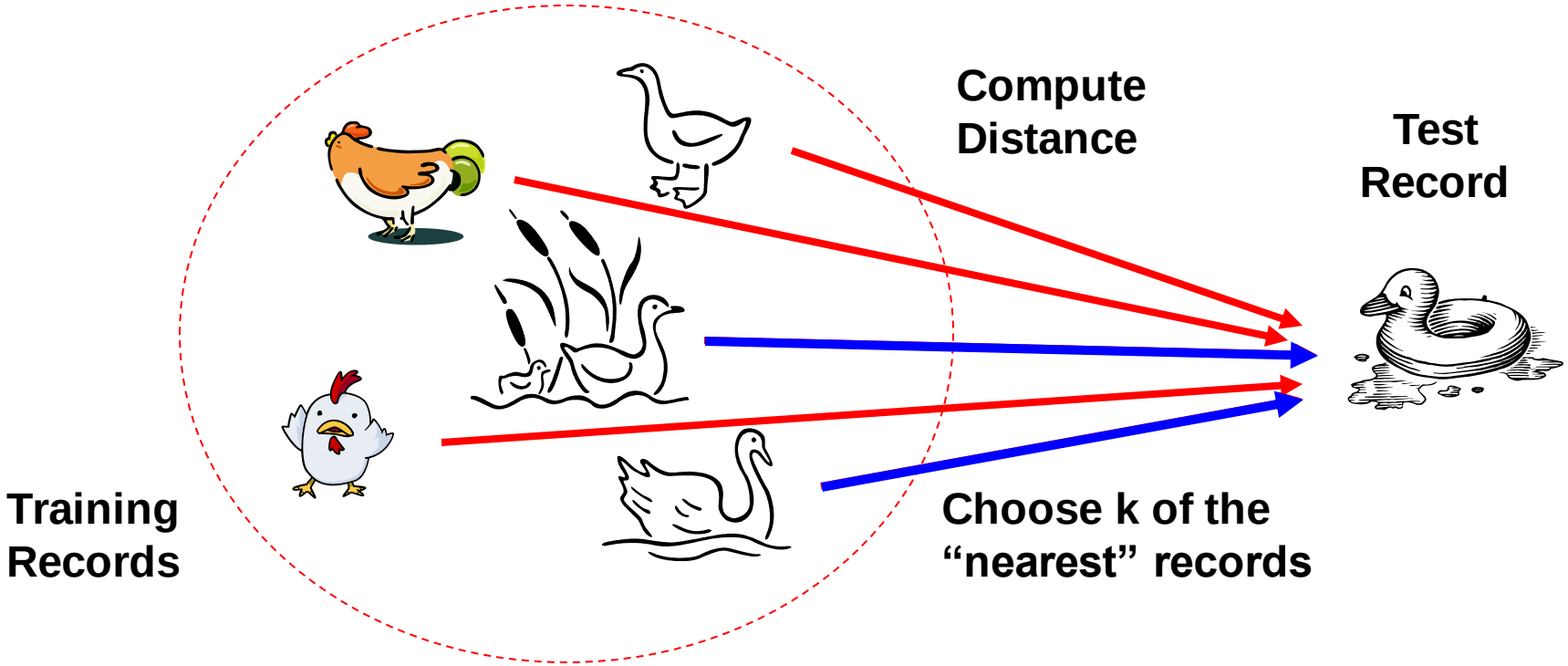
Introduction to Data Mining , 2nd Edition

by

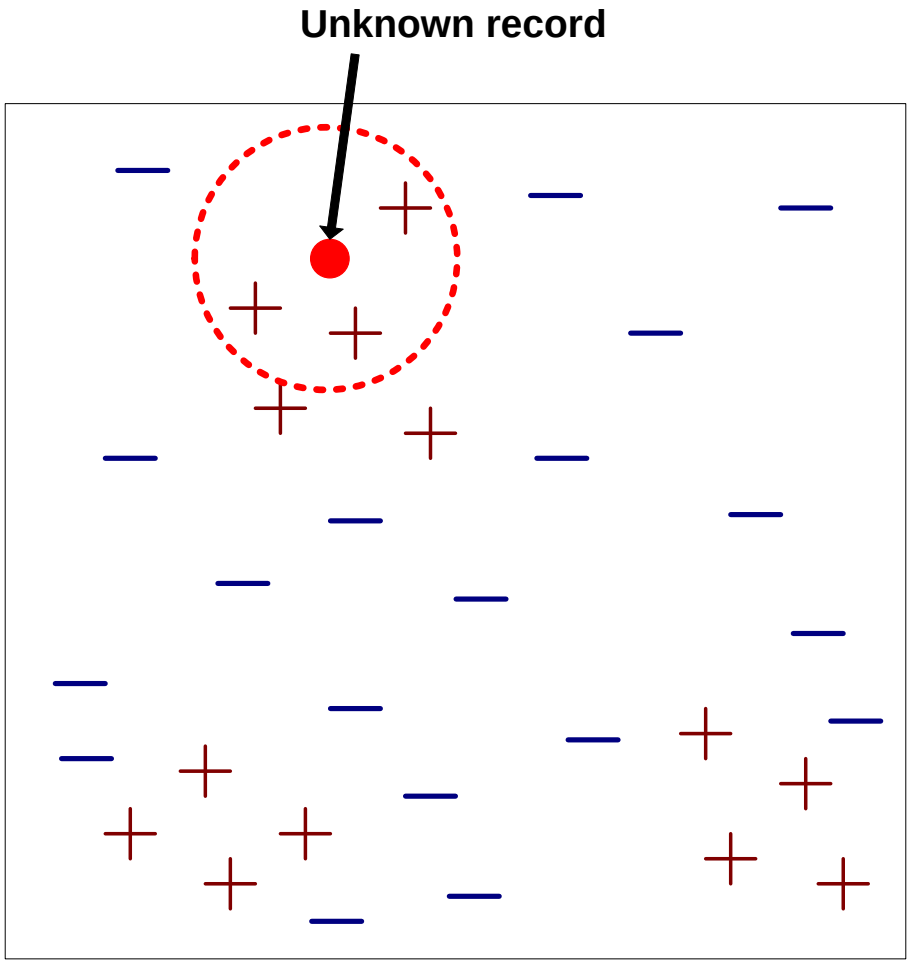
Tan, Steinbach, Karpatne, Kumar

Nearest Neighbor Classifiers

- Basic idea:
  - If it walks like a duck, quacks like a duck, then it's probably a duck



Nearest-Neighbor Classifiers



- Requires three things
  - The set of labeled records
  - Distance metric to compute distance between records
  - The value of *k*, the number of nearest neighbors to retrieve
- To classify an unknown record:
  - Compute distance to other training records
  - Identify *k* nearest neighbors
  - Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)

Nearest Neighbor Classification

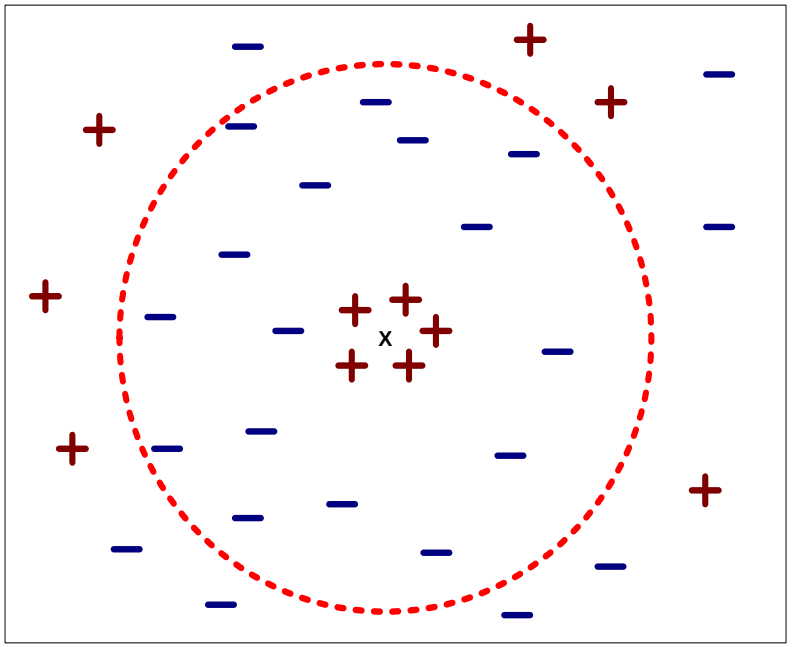
- Compute proximity between two points:
  - Example: Euclidean distance

$$d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{\sum_i (\boldsymbol{x_i} - \boldsymbol{y_i})^2}$$

- Determine the class from nearest neighbor list
  - Take the majority vote of class labels among the k-nearest neighbors
  - Weight the vote according to distance
    - weight factor, $w = 1/d^2$

Nearest Neighbor Classification…

- Choosing the value of k:
  - If k is too small, sensitive to noise points
  - If k is too large, neighborhood may include points from other classes



Nearest Neighbor Classification…

- **Choice of proximity measure matters**
  - For documents, cosine is better than correlation or Euclidean

1 1 1 1 1 1 1 1 1 1 1 0		0 0 0 0 0 0 0 0 0 0 0 1
	vs	
0 1 1 1 1 1 1 1 1 1 1 1		1 0 0 0 0 0 0 0 0 0 0 0

Euclidean distance = 1.4142 for both pairs

Nearest Neighbor Classification…

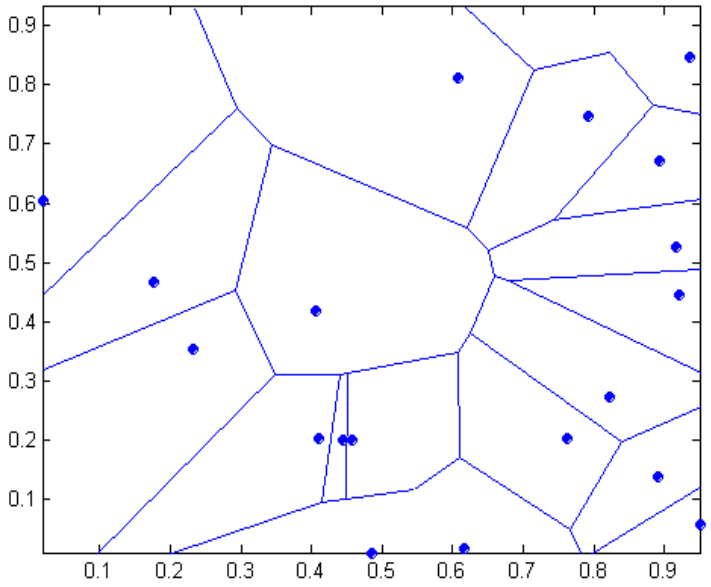
- **Data preprocessing is often required**
  - Attributes may have to be scaled to prevent distance measures from being dominated by one of the attributes
    - Example:
      - height of a person may vary from 1.5m to 1.8m
      - weight of a person may vary from 90lb to 300lb
      - income of a person may vary from $10K to $1M
  - Time series are often standardized to have 0 means a standard deviation of 1

Nearest-neighbor classifiers

- Nearest neighbor classifiers are local classifiers

- They can produce decision boundaries of arbitrary shapes.

1-nn decision boundary is a Voronoi Diagram



Nearest Neighbor Classification…

- **How to handle missing values in training and test sets?**
  - Proximity computations normally require the presence of all attributes
  - Some approaches use the subset of attributes present in two instances
    - This may not produce good results since it effectively uses different proximity measures for each pair of instances
    - Thus, proximities are not comparable

Nearest Neighbor Classification…

- **Handling irrelevant and redundant attributes**
  - Irrelevant attributes add noise to the proximity measure
  - Redundant attributes bias the proximity measure towards certain attributes
  - Can use variable selection or dimensionality reduction to address irrelevant and redundant attributes

Improving KNN Efficiency

- Avoid having to compute distance to all objects in the training set
  - Multi-dimensional access methods (k-d trees)
  - Fast approximate similarity search
  - Locality Sensitive Hashing (LSH)
- Condensing
  - Determine a smaller set of objects that give the same performance
- Editing
  - Remove objects to improve efficiency

Example 1

Example. Attributes: Age and Loan, both continuous. Class Default (Y and N) We should classify the last entry.

Both attributes are continuous, we will use Euclidian distance and illustrate classification with k=3.
If we do not standardize the data, the attribute Loan will have greater influence on the distance. The results in that case are given in the table. The entry will be classified with Y.

Age	Loan	Default	Distance	
25	$40,000	N	102000	
35	$60,000	N	82000	
45	$80,000	N	62000	
20	$20,000	N	122000	
35	$120,000	N	22000	2
52	$18,000	N	124000	
23	$95,000	Y	47000	
40	$62,000	Y	80000	
60	$100,000	Y	42000	3
48	$220,000	Y	78000	
33	$150,000	Y	8000	1
48	$142,000	?		

Euclidean Distance

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Example 2

The distance based on the standardized data is given in the table. We see that the neighbors of the entry have changed, and now it is classified as N.

Age	Loan	Default	Distance
0.125	0.11	N	0.7652
0.375	0.21	N	0.5200
0.625	0.31	N	0.3160
0	0.01	N	0.9245
0.375	0.50	N	0.3428
0.8	0.00	N	0.6220
0.075	0.38	Y	0.6669
0.5	0.22	Y	0.4437
1	0.41	Y	0.3650
0.7	1.00	Y	0.3861
0.325	0.65	Y	0.3771
0.7	**0.61**	**?**	

1 2 3

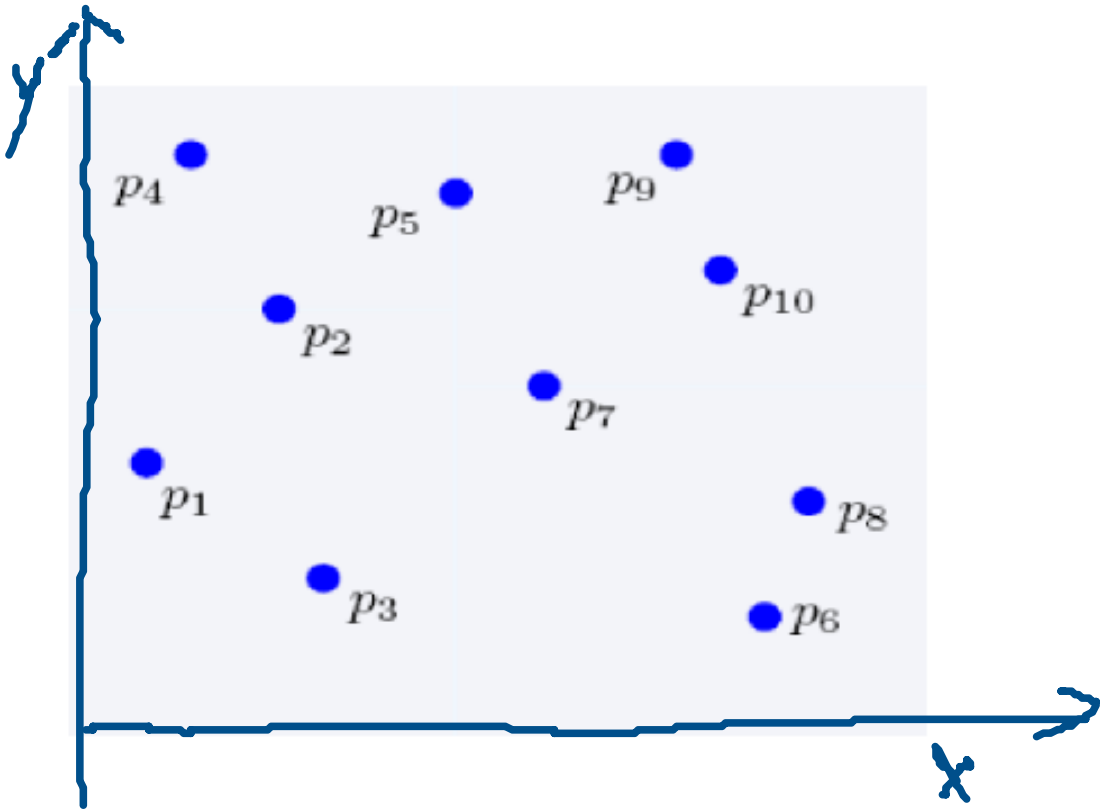
Standardized Variable

$$X_s = \frac{X - Min}{Max - Min}$$

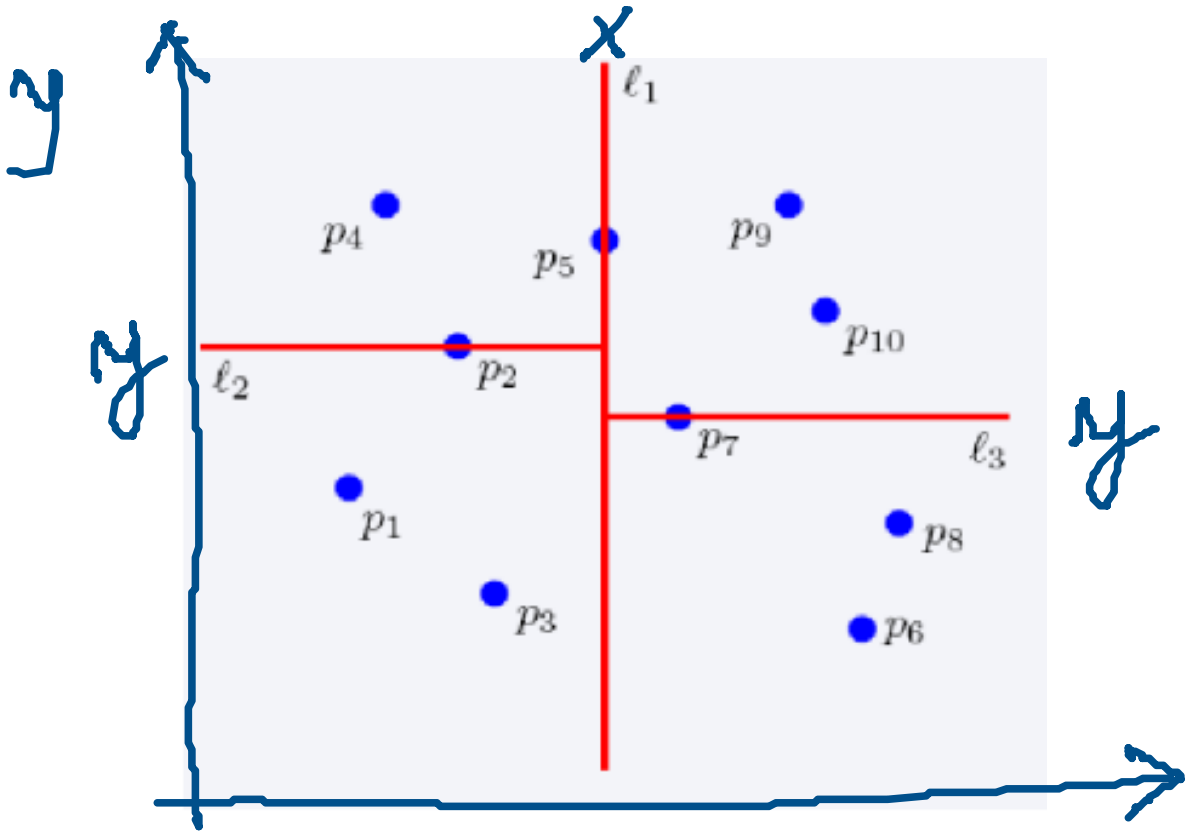
Mixed data

We should be careful when the data is mixed and Euclidean Hamming or other distances are not appropriate. The data may me transformed – continuous data may be standardized and categorized, or cosine metric can be used (with or without weights) to transform the data on (0,1) interval. However, there is no method which is efective on different mixed data sets. Research on different solutions – defining similarity/disimilarity measures, Gower distance, Heterogenous distance metric (ex. Heterogeneous Euclidean-Overlap Metric (HEOM) etc.)

Kd trees – approximation of the choice of knn

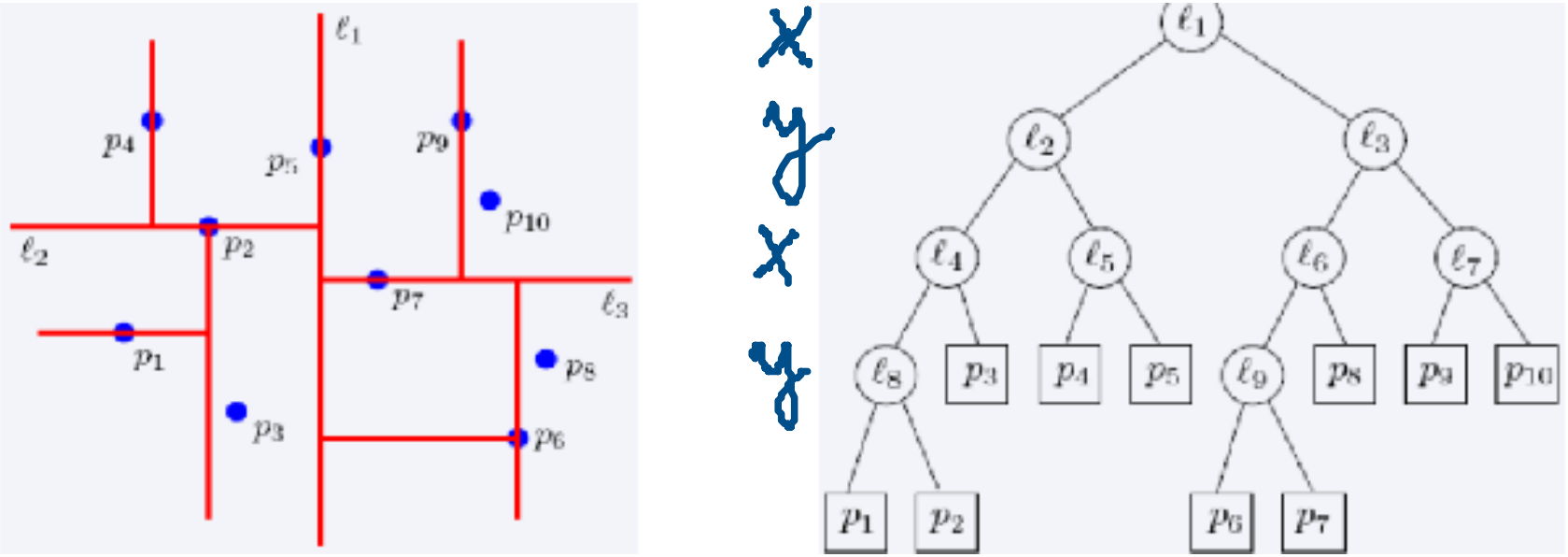


Kd trees



Separate by median or middle point of the range with respect to one attribute, x or y. The spliting point can be chosen to belong to the left or right subtree. At the next step make the splitin with respect to the other atribute etc. There are different techniques for the spliting criterion, depending of the application of the kd tree (example subinterval local vs global variance ratio when kd is used for outliers detection, etc.)

Kd trees



The tree can be constructed all the way down such that the leaves are the points of the data set as on the figure above. We can also stop at some predetermined number of points in a leaf (see next example). Such binary tree can help us find the closest neighbour of a point (or k neighbours).

Example 3.

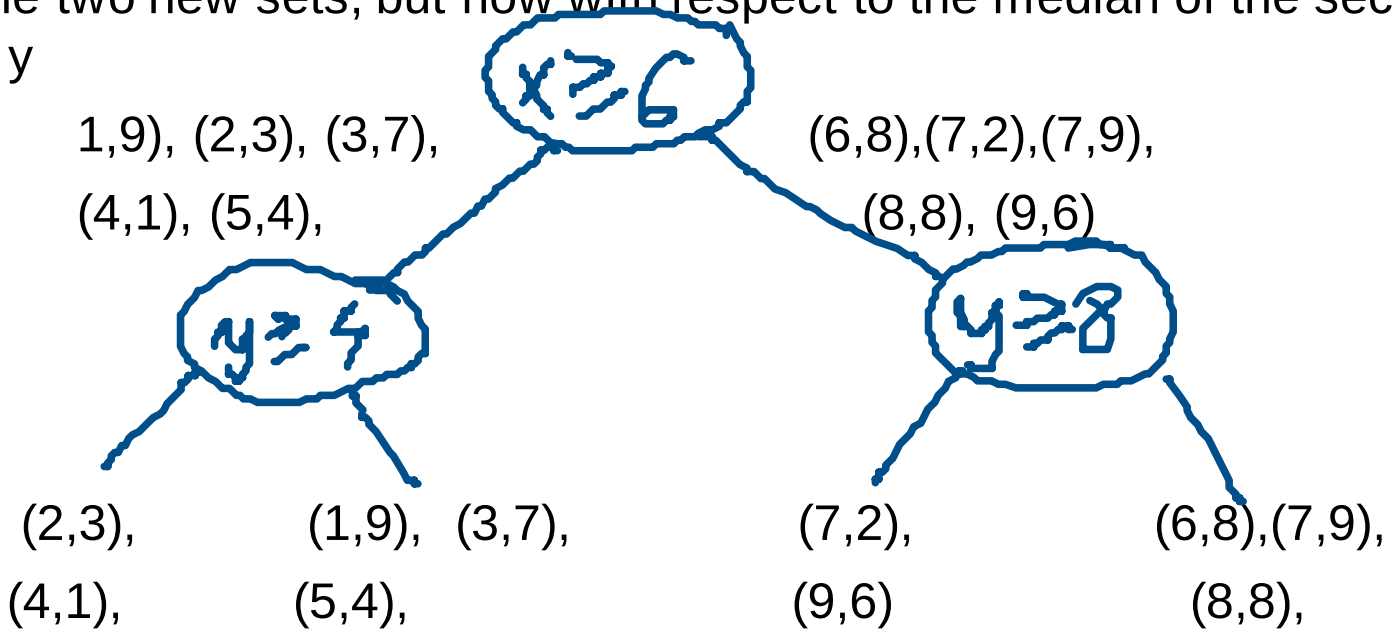
(x,y): (1,9), (2,3), (4,1), (3,7), (5,4), (6,8),(7,2), (8,8), (7,9), (9,6)

We can order the data by x (or by the compononet with greater variance)

(x,y): (1,9), (2,3), (3,7), (4,1), (5,4), (6,8),(7,2),(7,9), (8,8), (9,6)

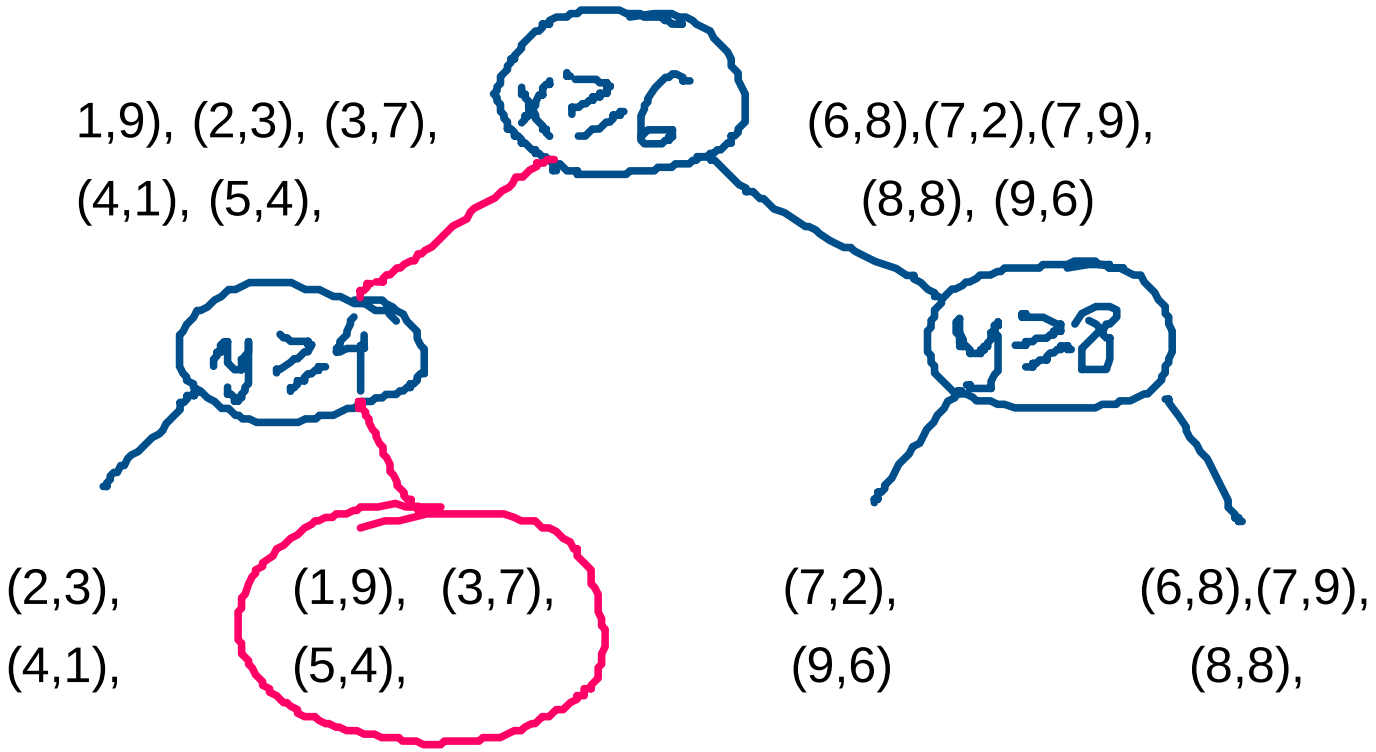
1. split the data at the point which is median for x atribute

2. split the two new sets, but now with respect to the median of the second attribute y



Example 3.

New point (2,7) - compare distances with (1,9), (3,7) and (5,4) and find the nearest neighbour. If we completely span the tree, the search will go all the way down to the leaves. We can use this procedure to find k nearest neighbours. Represent the points graphically and check the choice of



Example 4.

The data below has 3 atributes, for convinience the classes are not included. We first sort the data by column b which has the highest variance. If the data has low variance, we can assume that the point are more similar between each other. This is especially good approach if we use kd tree to detect outliers.

Table 0
Unsorted data

a	b	c
22	38	21
4	8	6
2	14	3
8	20	12
10	26	18
12	32	15
18	56	33
16	44	27
20	50	24
14	62	30
6	2	9

Table 1
Sort on column b

a	b	c
6	2	9
4	8	6
2	14	3
8	20	12
10	26	18
12	32	15 <--- median
22	38	21
16	44	27
20	50	24
18	56	33
14	62	30

Break *Table 1* on median (12,32,15)

Table 2

a	b	c
22	38	21
16	44	27
20	50	24
18	56	33
14	62	30

Table 3

a	b	c
6	2	9
4	8	6
2	14	3
8	20	12
10	26	18

Tables sorted on column C are as below.

Sort Table 2 on column c

Table 3

a	b	c
22	38	21
20	50	24
16	44	27 <--
14	62	30
18	56	33

Sort Table 3 on column c

Table 4

a	b	c
2	14	3
4	8	6
6	2	9 <--
8	20	12
10	26	18

med. med.

Table 3 is next split at median, (16,44,27), and Table 4 is split at median, (6,2,9), as below.

Break Table 3 on median (16,44,27)

	>			<	
14	62	30	22	38	21
18	56	33	20	50	24

Break Table 4 on median (6,2,9)

	>			<	
8	20	12	2	14	3
10	26	18	4	8	6

We can stop here in which case these four tables will be the leaves (draw the tree). Or we can continue by sorting column a, then split on b, c etc. Build the tree and at each node mark which is the spliting (comparing) attribute.

The data at the leaves are possible nearest points (though there may be a mistake). Sometimes we can choose to make more comparisons, using the tables (data) on the other side of the last parent node. We can also go one level up and make more comparisons.